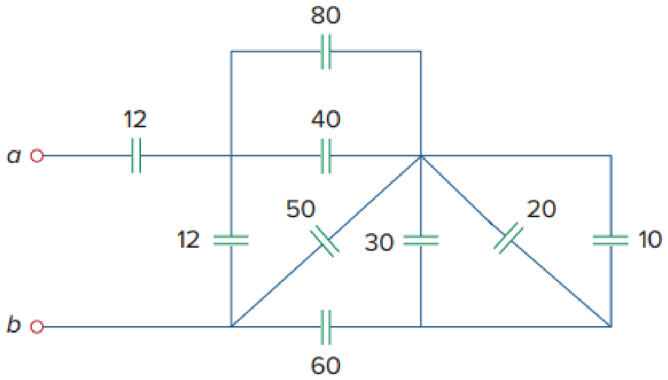
Chapter 6, Problem 19P

Problem

Find the equivalent capacitance between terminals *a* and *b* in the circuit of Fig. 6.53. All capacitances are in μF.

Fig. 6.53:



Step-by-step solution

Step 1 of 6

Refer to figure 6.53 given in the textbook.

The capacitances $10\ \mu\text{F}$, $20\ \mu\text{F}$, and $30\ \mu\text{F}$ are parallel to each other.

Calculate its equivalent C_1.

$$\begin{aligned} C_1 &= 10+20+30 \\ &= 60\ \mu\text{F} \end{aligned}$$

Redraw the circuit as shown in figure 1.

Consider that all capacitances in μF.

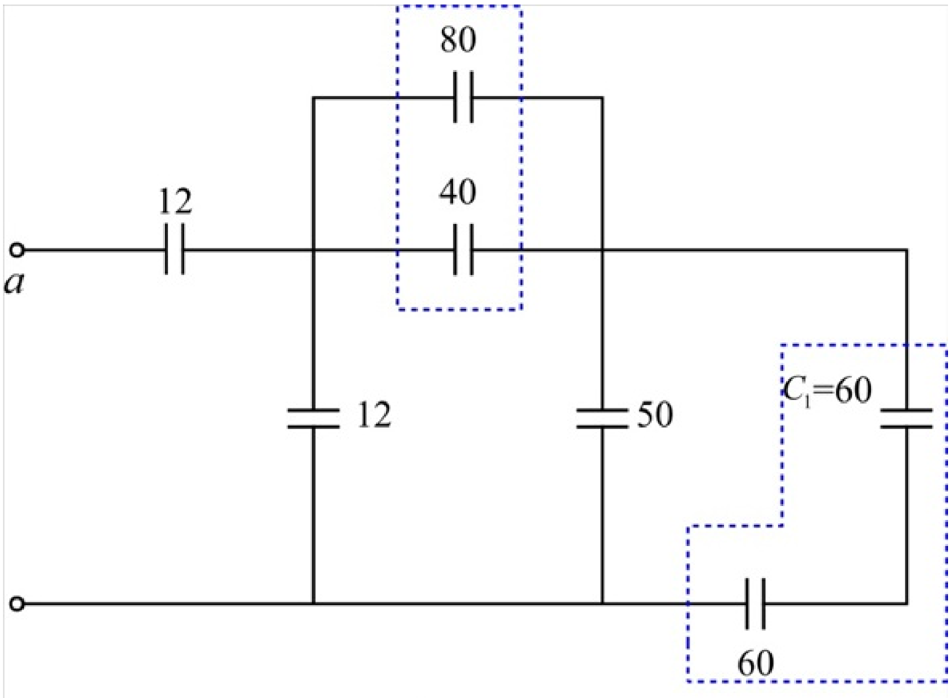


Figure 1

Step 2 of 6

Consider the circuit shown in figure 1.

Calculate the equivalent capacitance of the capacitors $80\ \mu\text{F}$ and $40\ \mu\text{F}$.

They are parallel to each other.

$$\begin{aligned} C_2 &= 80+40 \\ &= 120\ \mu\text{F} \end{aligned}$$

Calculate the equivalent capacitance of the capacitors $60\ \mu\text{F}$ and $60\ \mu\text{F}$.

They are parallel to each other.

$$\begin{aligned} C_3 &= \frac{60\times 60}{60+60} \\ &= \frac{3600}{120} \\ &= 30\ \mu\text{F} \end{aligned}$$

Step 3 of 6

Redraw the network with the equivalent capacitances C_2, and C_3 as shown in figure 2.

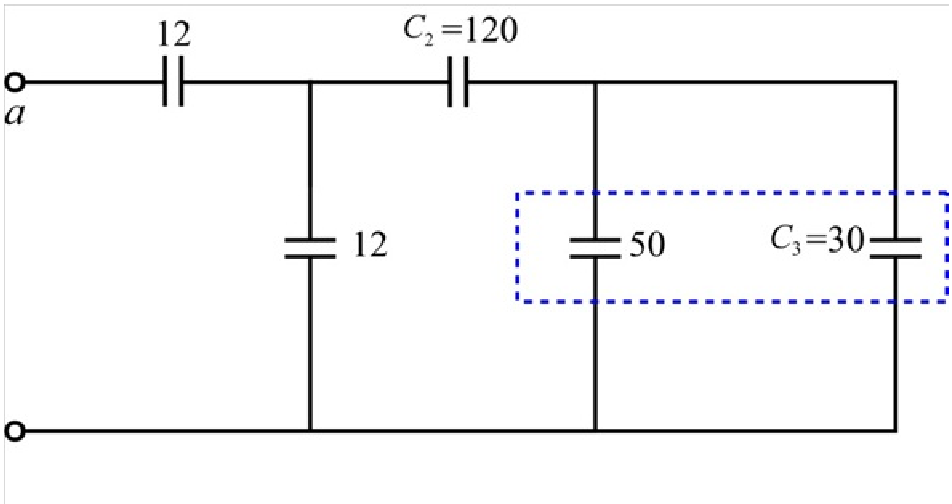


Figure 2

Step 4 of 6

Consider the circuit shown in figure 2.

Calculate the equivalent capacitance of the capacitors $50\ \mu\text{F}$ and $30\ \mu\text{F}$.

They are parallel to each other.

$$\begin{aligned} C_4 &= 50+30 \\ &= 80\ \mu\text{F} \end{aligned}$$

The capacitances C_4 and C_2 are in series.

The equivalent capacitance is,

$$C_5 = \frac{C_2 \times C_4}{C_2 + C_4}$$

Substitute 80 for C_4 and 120 for C_2.

$$\begin{aligned} C_5 &= \frac{120 \times 80}{120+80} \\ &= \frac{120 \times 80}{200} \\ &= \frac{12 \times 8}{2} \\ &= 48\ \mu\text{F} \end{aligned}$$

Step 5 of 6

Redraw the circuit with equivalent capacitance C_5 as shown in figure 6.

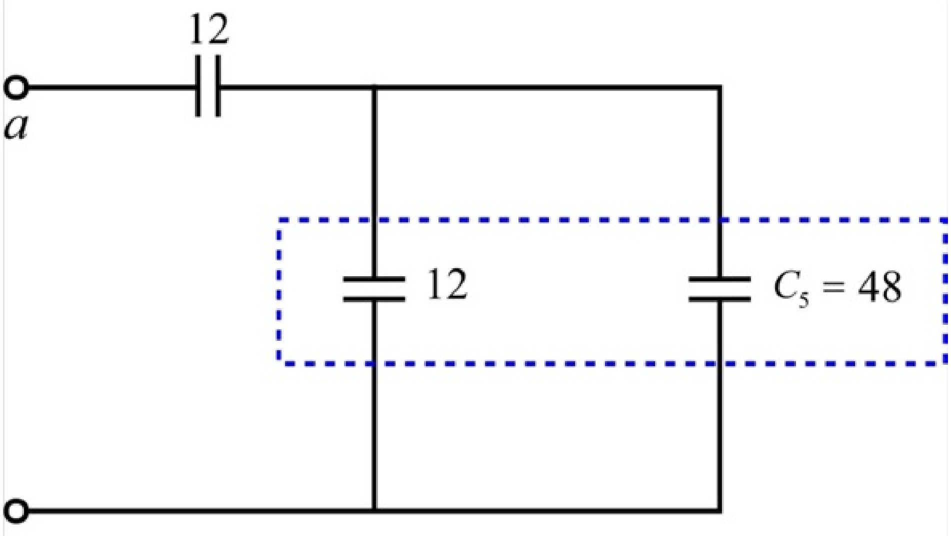


Figure 3

Step 6 of 6

Consider the circuit shown in figure 3.

Calculate the equivalent capacitance of the capacitors $12\ \mu\text{F}$ and $48\ \mu\text{F}$.

They are parallel to each other.

$$\begin{aligned} C_6 &= 48 + 12 \\ &= 60\ \mu\text{F} \end{aligned}$$

The capacitances $12\ \mu\text{F}$ and C_6 are in series.

The equivalent capacitance is,

$$C_5 = \frac{12 \times C_6}{12 + C_6}$$

Substitute 60 for C_6.

$$\begin{aligned} C_{eq} &= \frac{12 \times 60}{12 + 60} \\ &= \frac{720}{72} \\ &= 10\ \mu\text{F} \end{aligned}$$

Therefore, the equivalent capacitance between terminals *a* and *b* is $\boxed{10\ \mu\text{F}}$.